

NorthBay Pantry Enterprise Data Platform

Designing an Enterprise Data Platform from Scratch

DSCI 504 · UBA MADS · Ahmad Var · May 2026

The Problem — \$1.4B in Revenue, Zero Data Infrastructure

160 stores across US + UK · E-commerce · Pantry Box meal-kit · PantryPlus loyalty

10 data sources:

- Some streaming at 80,000 events/minute
- Some nightly CSV files with garbled headers and encoding issues
- Some WAV audio files from 250 hours/day of customer support calls

Finance can't see revenue until spreadsheets are emailed each morning

Customer Success doesn't know a caller's order history when they pick up the phone

Data Science can't train models — no consistent, centralized data access

→ This is a greenfield build at \$1.4B enterprise scale. No data warehouse. No data team.

Design Philosophy — Match Tool to Workload

A single database cannot serve real-time streaming, sub-second dashboards, and 5-year model training simultaneously.

Three questions drove every design decision:

1. What is the shape of the data? Structured / semi-structured / binary (audio, HTML)
2. What is the access pattern? Streaming / batch / sub-second online lookup
3. Who owns the data and who can see it?

Engineering decisions are principled — not arbitrary.

Every technology choice traces back to one of these three questions.

D1 — Reference Architecture: Bronze → Silver → Gold

10 Sources → Ingestion Layer → Bronze (GCS) → Silver (BigQuery) → Gold (BigQuery) → Consumers

↑ ↑ ↑
Raw, immutable Cleaned, validated Business-ready
never modified bad rows quarantined role-gated

Bronze (GCS)	Raw files, forever. The recovery layer — any downstream bug is fixed by reprocessing from here.
Silver (BigQuery)	dbt cleans, validates, deduplicates. Entity resolution produces one customer_uuid per person.
Gold (BigQuery)	Finance, Marketing, and Data Science query here. Access enforced by role.
Feature Store	Feast + Redis — ML features served in <5ms to meet the Customer Success 2-second SLA.

Orchestration: Apache Airflow (batch pipelines) + Apache Kafka (streaming events)

D2 — Ingestion: 10 Sources, 3 Patterns

STREAMING (real-time)

Sources: OLTP PostgreSQL (CDC), Mobile App Events, MongoDB Change Streams

Tool: Debezium → Kafka → Kafka Connect → Bronze

BATCH (scheduled)

Sources: Store POS CSV/SFTP (160 stores), Salesforce, PantryPlus Loyalty, 4 Marketing Platforms, Supplier EDI

Tool: Apache Airflow — one DAG per source

EVENT-DRIVEN (continuous)

Sources: Customer Support Call WAV files (~250 hrs/day)

Tool: S3 event → SQS → Lambda → AWS Transcribe → BigQuery

KEY DESIGN DECISIONS

Fault isolation on POS One Airflow task per store. One broken CSV → quarantine + alert. 159 stores continue unaffected.

Idempotency everywhere Bronze partitioned by source/YYYY/MM/DD/run_id — re-runs overwrite the same partition, never double-count.

Rate limiting PantryPlus API capped at 60 req/min via token-bucket limiter built into the Airflow operator.

D3 — Storage Architecture: Right Tool for the Right Workload

- GCS / S3 Raw files, WAV audio, scraped HTML
Cheapest \$/GB · no schema enforcement · handles any binary format
 - Apache Kafka In-flight events from OLTP, Mobile App, MongoDB
Decouples producers from consumers · replay on failure · Schema Registry enforces Avro contracts
 - BigQuery (Silver/Gold) Cleaned tables, dashboards, all analytical queries
Columnar · TB-scale · sub-second aggregations · native dbt integration
 - Feast + Redis Online ML feature store — <5ms feature lookup for Customer Success
BigQuery's ~1-3 sec query latency violates the 2-second SLA. Redis solves it.
 - Feast + BigQuery Offline ML feature store — full history for training churn, demand forecast, recommender
- "Why not one system?"
- BigQuery can't store WAV files. S3 can't answer `SELECT SUM(revenue) GROUP BY store` in milliseconds.
 - Redis exists for one specific reason: to serve a single SLA that BigQuery cannot meet.

D4 — Curation: The Customer Identity Problem

Same person — 6 different records across 6 systems:

"ahmad.var@gmail.com" in OLTP PostgreSQL
"AHMAD VAR" + phone in Salesforce
"ahmadvar@gmail.com" in PantryPlus loyalty (typo variant)
Device ID only in Mobile App
Hashed email in Marketing platforms
Username (pseudonymous) in Customer Reviews

Entity Resolution Pipeline:

- 1 Deterministic pass Exact match on normalized email → lower().strip() → same customer, guaranteed
- 2 Probabilistic pass Splink scoring: name + phone Soundex + address zip ≥ 0.85 → merge candidate
- 3 Golden record One customer_uuid (UUID v4) written to golden_customer_master
- 4 Survivorship rule Most recently updated non-null field wins · OLTP email is canonical

Also:

golden_product_master MongoDB is system of record · flattens nested attrs · standardizes FALCPA allergen codes
All amounts → USD Daily ECB rate · original currency stored alongside
All timestamps → UTC Display timezone applied at the query/API layer only

D5 — Data Quality: Bad Data Has a Home, But Not in Production

Schema conformance	Bronze	· Great Expectations	→ Quarantine file · Slack #data-alerts
Null rate on required fields	Silver	· dbt not_null	→ Block Silver load · PagerDuty P2
Duplicate primary keys	Silver	· dbt unique	→ Block load · alert
Row count $\pm 20\%$ vs prior day	Silver	· dbt custom	→ PagerDuty P1 if Finance table
Referential integrity (FK)	Silver	· dbt relationships	→ Log violation · do not block
PII in non-PII table	Gold	· GE custom expectation	→ Block + incident ticket
Finance table immutability	Gold	· audit trigger	→ Block + SOX violation alert
Freshness SLA miss	Gold	· Airflow SLA callback	→ PagerDuty P1 if 7AM Finance miss

Quarantined rows land in bq_quarantine with reason code + timestamp.
Never deleted — reviewed weekly by data stewards.

Results surfaced in DataHub data quality dashboard, visible to all owners and stewards.

D6 + D7 — Catalog and Governance

DataHub (open-source catalog)

Every column tagged: domain · sensitivity (public / internal / confidential / restricted) · pii · source_system · owner

Source-to-Gold lineage auto-ingested from dbt — full column-level lineage without manual effort

PII tags flow: dbt meta:{pii:true} → DataHub → BigQuery column-level policy tag enforcement

Governance — 3 Roles, Clear Accountability

Data Owner (VP) Accountable for accuracy and access policy for their domain

Data Steward (Analyst) Reviews quarantine queue · approves schema change requests

Data Custodian (Data Eng) Implements access controls · runs and monitors pipelines

RACI Highlight

golden_customer_master Owner: VP Customer Success | Steward: CS Analyst | Legal: Consulted

finance_daily_revenue Owner: CFO | Steward: Finance | Legal: Informed

stg_call_transcripts Owner: VP Customer Success | Steward: CS Analyst | Legal: ACCOUNTABLE

Voice recordings are the only dataset where Legal is Accountable — not just Informed.

D8 — Security and Compliance: Two Frameworks, One Platform

Access Controls

BigQuery column-level policy tags: call transcripts readable only by role:customer_success

AES-256 at rest on all GCS + BigQuery · WAV files use CMEK (customer-managed keys) · TLS 1.3 in transit

Cloud Audit Logs retained 7 years (SOX)

SOX tables (finance_daily_revenue): append-only · no UPDATE/DELETE · change-controlled via dbt PR approval

GDPR Right-to-Erasure Runbook — target 5 business days, hard limit 30 days

- 1 Look up customer_uuid in golden_customer_master using email or phone
- 2 DataHub lineage API → lists every table containing customer_uuid FK — in seconds, not hours
- 3 Erasure dbt macro: nullify PII columns in Silver → [ERASED-{date}]
- 4 Delete WAV files + transcripts from S3/BigQuery using call UUID index
- 5 Submit deletion requests to Salesforce API + PantryPlus SaaS vendor
- 6 Bronze: PII stays (immutable raw) — permissible under GDPR audit exception · encrypted + access-restricted
- 7 Log completion in gdpr_erasure_log · notify customer within 30-day SLA

The DataHub lineage answer to "find every row about customer X" takes seconds, not hours.

D9 — Monitoring and Operations

Pipeline SLAs

Finance daily revenue	Available by 07:00 ET	P1 — page on-call immediately
Store POS ingestion	Complete by 05:00 ET	P1 — Finance SLA depends on it
Silver refresh	Complete by 06:00 ET	P1
Customer 360 lookup	< 2 sec response	P2 — page within 15 min
Loyalty / Marketing	Complete by 09:00 ET	P2

Alerting Chain

Airflow SLA miss callback → PagerDuty (P1 or P2) → On-call data engineer → Slack #data-incidents

On-call: Weekly rotation among 3 engineers. Each DAG has a `runbook_url` tag — one click to the runbook.

Key Metrics (Airflow + Datadog)

DAG success rate · Bronze→Silver lag (minutes) · quarantine rate per source
Kafka consumer lag · BigQuery slot utilization + cost per DAG · Gold table freshness

Cost tagging on every job: `team / source_system / environment / dag_id` → monthly VP-level cost reports

D10 — Sample Artifacts: The Design Is Backed by Working Code

`artifacts/dags/pos_ingestion_dag.py`

Per-store fault isolation via try/except — one broken file cannot stop the other 159 stores
chardet for encoding detection · quarantine bucket + Slack alert on failure

`artifacts/dbt/stg_pos_transactions.sql`

Bronze → Silver transformation: `SAFE_CAST` on every column · dedup via `row_number()`
Incremental MERGE on natural key — guarantees idempotency on any re-run

`artifacts/great_expectations/pos_suite.py`

DQ checks: nulls on required fields · unique transaction IDs · revenue in valid range
Row count freshness check — fails if today's batch is missing entirely

`artifacts/dags/alert_callback.py`

SLA miss callback: fires PagerDuty P1 payload + posts to Slack `#data-incidents` automatically
Includes last-success timestamp in the alert so on-call engineer has immediate context

Each artifact demonstrates a different layer of the platform.

The per-store try/except in the DAG is the most readable illustration of fault isolation.

D11 — Cost and Risk: ~\$20K-\$33K/month

Component	Monthly Estimate
BigQuery (storage + queries)	\$4,000 – \$8,000
GCS / S3 (Bronze + audio)	\$2,000 – \$3,500
Kafka (Confluent Cloud)	\$3,000 – \$5,000
Airflow (Cloud Composer)	\$2,000 – \$3,000
AWS Transcribe (250 hrs/day)	\$8,000 – \$12,000 ← DOMINANT COST (40% of total)
DataHub + Redis	\$1,000 – \$2,000
TOTAL	~\$20,000 – \$33,500/month

Mitigation: Transcribe only escalated calls → reduce audio cost by ~60%

Top 3 Risks

HIGH / HIGH POS schema drift breaks Silver
→ Schema evolution tests in CI · unknown columns quarantined on arrival

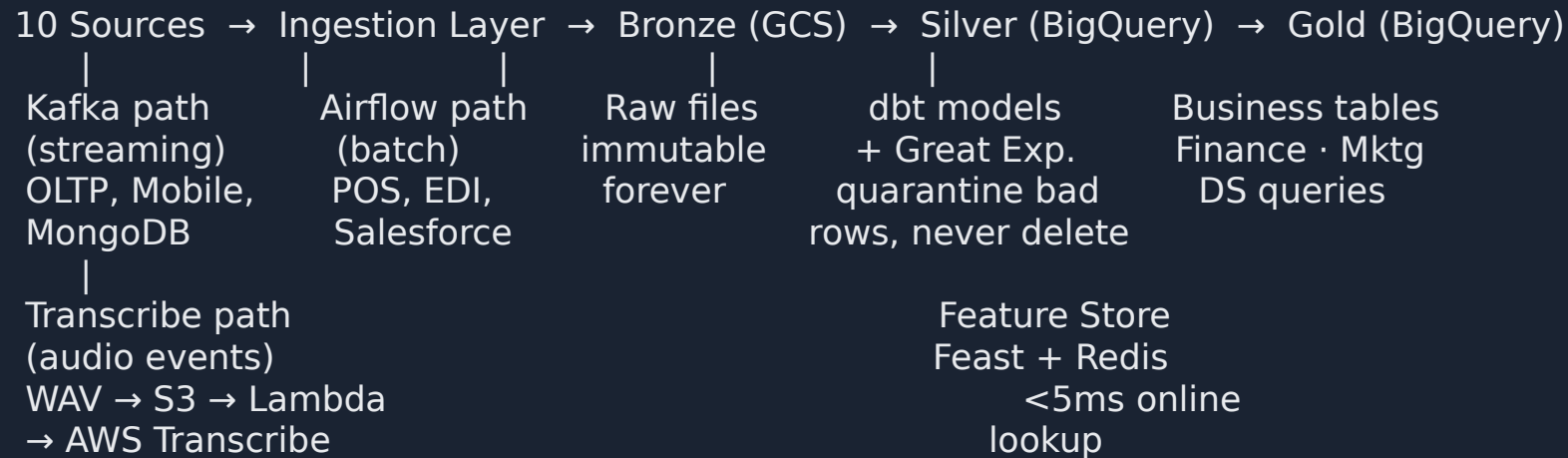
MEDIUM / MEDIUM Kafka consumer lag during mobile event spike
→ Auto-scaling consumer group · 7-day topic retention for replay

LOW / CRITICAL GDPR erasure misses a table
→ DataHub lineage + quarterly erasure drill + Legal audit of erasure log

8 Concrete Questions — Rapid Fire

- 1 How does data flow from each source?
D2 ingestion table — 3 patterns, 10 sources, specific tool per source
- 2 Where does each dataset live and why?
D3 storage table — each system exists because no other system can do its job
- 3 How do you prevent double-counting on re-runs?
Bronze overwrite partitions (source/YYYY/MM/DD/run_id) + Silver MERGE on natural key
- 4 How do you identify one customer across 10 sources?
customer_uuid via deterministic email match first, then Splink probabilistic scoring (≥ 0.85)
- 5 How do you find every row about customer X in under one hour for GDPR?
DataHub lineage API lists all tables with customer_uuid FK in seconds → erasure macro runs in minutes
- 6 Which datasets are most sensitive and how is access controlled?
Support call transcripts (Restricted) + customer PII (Confidential) — BigQuery column-level policy tags
- 7 If Finance dashboard is missing at 7 AM — what happens?
Airflow SLA miss callback → PagerDuty P1 → on-call engineer with runbook URL in the alert body
- 8 What would this cost to run?
~\$20K–\$33K/month · audio transcription is the cost lever (40% of total, reducible by ~60%)

Architecture Recap — End to End



Key paths:

Real-time events → Kafka → Kafka Connect → Bronze → Silver streaming model
Daily batches → Airflow DAGs (one per source) → Bronze → dbt Silver → Gold
Audio calls → S3 event trigger → Lambda → AWS Transcribe → BigQuery

NorthBay — From Zero Infrastructure to Enterprise-Grade Platform

- ✓ 10 sources ingested reliably and idempotently
- ✓ One trusted customer identity across all systems
- ✓ Finance gets their 7 AM dashboard — automated runbook fires if it's ever late
- ✓ Data Science has a feature store with sub-millisecond online serving
- ✓ Legal has a GDPR runbook that completes in 5 business days, not weeks
- ✓ ~\$20K-\$33K/month — with a clear cost lever on audio transcription

Every design decision is auditable.

Every dataset has an owner.

Every alert has a runbook.

The platform is not just built — it's operable.